

Curriculum Vitae/Resume

Rupadarshi Ray

rupadarshiray.github.io

Master of mathematics (awaited)
IISER Mohali

rupadarshiray@gmail.com

Thesis and interests

My primary interests lie in the topics related to the analysis of manifolds: be it geometric, topological, dynamical or analytical. I have invested a lot of time on Riemann surfaces, hyperbolic manifolds and discrete subgroups of $\text{Isom}(\mathbb{R}H^n) \cong O^+(1, n)$ during my thesis:

[Quasiconformal and measurable rigidity of discrete groups](#) Fall 2025-Spring 2026
Masters thesis under Dr. Arghya Mondal IISER Mohali

- ★ Tukia's quasi-conformal rigidity of discrete groups of $\text{Isom}(\mathbb{R}H^n) \cong O^+(1, n)$ that diverge at $n - 1$
- ★ Patterson-Sullivan densities of discrete subgroups of $\text{Isom}(\mathbb{R}H^n) \cong O^+(1, n)$
- ★ Sullivan's measurable rigidity theorem on discrete subgroups of $\text{Isom}(\mathbb{R}H^n) \cong O^+(1, n)$ of divergent type

My broad interests also include geometric theory of holomorphic functions/quasiconformal mappings, harmonic analysis, ergodic theory, symplectic geometry and Hamiltonian flows, representation theory of Lie groups and Lie algebras, etc.

Academic referees

- Arghya Mondal, arghyamondal@iisermohali.ac.in, Assistant Professor, IISER Mohali (MS thesis supervisor)
- Kapil Hari Paranjape, kapil@iisermohali.ac.in, Professor Emeritus, IISER Mohali
- Vaibhav Vaish, vaibhav@iisermohali.ac.in, Assistant Professor, IISER Mohali
- Pranab Sardar, psardar@iisermohali.ac.in, Assistant Professor, IISER Mohali
- Soma Maity, somamaity@iisermohali.ac.in, Assistant Professor, IISER Mohali

Education

[Indian Institute of Science Education and Research Mohali \(IISER Mohali\)](#) 2021-2026
5 years BSMS, graduating with a Masters in Mathematics Mohali, Punjab, India

- ★ [Graduating CPI: 8.6](#)
- ★ [Degree obtained on June 16, 2026](#)

Techno India Group Public School (TIGPS), Balurghat 2007-2020
Primary, Middle and High School Balurghat, West Bengal, India

- ★ Standard XII CBSE board exams: 93.6%
- ★ Standard X CBSE board exams: 90.6%

Background and involvement

Beyond the standard curriculum for masters, I have also taken the following courses.

2022 *Geometry of curves and surfaces* following *Pressley and do Carmo*
 2023 Audited the course on *algebraic topology* following *Hatcher*
 2024 *Knots and braids*: Reidemeister moves, colouring invariants, Jones polynomial, Alexander module and its elementary ideals, braid groups
 2024 *Riemannian geometry* following *Jack Lee*
 2025 *Arithmetic of elliptic curves*
 2025 *Fourier analysis*
 2026 *Fuchsian groups*

After self-studying the theory of smooth manifolds and differential forms I went on to reading the following during the summers.

Sheaf theoretic proof of de Rham isomorphism Summer 2023
Summer reading under Dr. Shane D' Mello IISER Mohali

- ★ *Jack Lee* on de Rham cohomology of differential forms
- ★ *Griffiths and Harris* on sheaf cohomology of differential forms
- ★ *Bott and Tu* on Cech-de Rham double complex

Representation theory of groups and Lie algebras Winter 2023 - Summer 2024
Reading under Dr. Vaibhav Vaish IISER Mohali

- ★ *Fulton and Harris* on representation theory of finite groups and Lie algebras
- ★ *Woit* on quantum mechanics and representation theory

Ergodic theorems for actions of locally compact groups Summer 2025
Summer reading under Dr. Jotsaroop Kaur IISER Mohali

- ★ Following *Einsiedler and Ward*, I presented proofs of ergodic theorems for measure preserving transformations and actions of Amenable groups.

Seminars and workshops that I have attended/participated in

[Geometric Aspect of Algebraic Varieties](#) Spring 2025
Conference and celebration of mathematical legacy of Dr. Kapil Hari Paranjape IISER Mohali

Meromorphic functions on Riemann surfaces Summer 2025
Seminar organized by self IISER Mohali

- ★ Talk by Dr. Kapil Hari Paranjape on etale space of sheaf of holomorphic and meromorphic functions, ringed space definition of Riemann surfaces
- ★ Student talks on constructing meromorphic functions on Riemann surfaces, Riemann-Roch theorem, Abel's theorem and Jacobi's inversion theorem

[Rigidity of Discrete Groups](#) June 30 - July 4, 2025
Workshop organized by Dr. Krishnendu Gongopadhyay and Dr. Pranab Sardar IISER Mohali

- ★ Attended talks by Dr. Marc Bourdon, Dr. C S Rajan
- ★ Attended talks on the Mostow rigidity theorem for hyperbolic 3-manifolds by Dr. Arghya Mondal

[Singapore-Bangalore "What is ...?" Workshop on Mathematics](#) Fall 2025
Workshop organized by NUS Singapore and IISc Bangalore IISc Bangalore

Boundary of symmetric spaces Fall 2025
Seminar organized by Dr. Arghya Mondal IISER Mohali

- ★ Symmetric spaces of non-compact type, $P(n, \mathbb{R})$ and its totally geodesic submanifolds
- ★ Boundary of CAT(0) symmetric spaces

Talks that I have given

- 2026 A talk on computing the **genus of modular curves**, for the course on *Fuchsian groups*, IISER Mohali, Spring 2026
- 2026 Two talks on the proof of **Sullivan's measure rigidity theorem**, *Boundary of symmetric spaces* seminar, IISER Mohali, Spring 2026
- 2026 An elementary proof of **Selberg's lemma for $SL(2, \mathbb{C})$** , for the course on *Fuchsian groups*, IISER Mohali, Spring 2026
- 2026 Multiple talks on the construction of **Patterson-Sullivan measures on limit sets**, *Boundary of symmetric spaces* seminar, IISER Mohali, Spring 2026
- 2026 An elementary talk on **proper actions and properly discontinuous actions** on topological spaces
- 2025 An introductory talk on **deformations and rigidity** of geometric structures: Teichmuller spaces and Mostow's rigidity following [an article by Lizhen Ji](#)
- 2025 **Reductive subgroups of $GL(n, \mathbb{R})$ and totally geodesic submanifolds of $P(n, \mathbb{R})$** : *Boundary of symmetric spaces* seminar, IISER Mohali, Fall 2025
- 2025 A talk on a sketch of the proof of **Gelfand-Naimark theorem** for C^* -algebras, for a series of talks on *harmonic analysis*
- 2025 Multiple talks on the proof of **Ergodic theorems for actions of locally amenable compact groups**, as presentations for a summer reading, IISER Mohali, Summer 2025
- 2025 **Abel's theorem and Jacobi's inversion theorem**: for the seminar on *Meromorphic functions on Riemann surfaces*, IISER Mohali, Summer 2025
- 2025 An introductory talk on **symplectic manifolds and Hamiltonian flows** titled "[When does a vector field kill area?](#)" in the [Graduate Students Seminar, IISER Mohali](#), Spring 2025
- 2025 **Construction of Haar measure on locally compact Hausdorff groups**: for the course on Fourier analysis, IISER Mohali, Semester 8
- 2025 A short talk on **construction of Riemann surface of holomorphic functions** for the course on seminar delivery, IISER Mohali, Semester 8
- 2025 **Periods of elliptic curves**: for the course on Arithmetic of elliptic curves, IISER Mohali, Semester 8
- 2024 A talk on **Knower's and liar's paradoxes and their formalizations** titled "What does a knower and a liar have in common?", course on Philosophy of language, IISER Mohali, Semester 7
- 2024 A short talk on **topological conjugacy of linear flows** for the course on seminar delivery, IISER Mohali, Semester 7
- 2024 A talk on **some comparison theorems in Riemannian geometry** for the course on Riemannian geometry, IISER Mohali, Semester 7

Skills

- English [TOEFL iBT \(November 2025\)](#) scored 99/120 (R: 25, L: 28, S: 24, W: 22)
- Teaching I have taught physics and mathematics to students at the undergraduate level, in particular for courses such as linear algebra, geometry of curves and surfaces, ordinary differential equations.
- Python Experience in NumPy, Sympy and Matplotlib for data analysis, algebra and visualization. Also used libraries such as SnapPy and Sage.
- L^AT_EX Proficient in creating complex documents, including academic papers and presentations.
- Notetaking I use [Obsidian](#) for taking notes on mathematics. I have experience in organizing and structuring information effectively, spread over hundreds of notes, using this tool.
- Vector art 10 years of experience in vector art design using Adobe Illustrator
- Other Familiar with C, C++, MATLAB, R, HTML, CSS, Javascript, Adobe Photoshop